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A statistical sensing method by utilizing Wi-Fi CSI subcarriers: Empirical study and performance enhancement

Tao Deng^{a,1}, Bowen Zheng^{b,1}, Rui Du^a, Fan Liu^{b,*}, Tony Xiao Han^a^a Wireless Technology Laboratory, Huawei Technologies Co., Ltd., Shenzhen 518000, China^b School of System Design and Intelligent Manufacturing (SDIM), Southern University of Science and Technology, Shenzhen 518000, China

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ABSTRACT

In modern Wi-Fi systems, channel state information (CSI) serves as a foundational support for various sensing applications. Currently, existing CSI-based techniques exhibit limitations in terms of environmental adaptability. As such, optimizing the utilization of subcarrier CSI stands as a critical avenue for enhancing sensing performance. Within the OFDM communication framework, this work derives sensing outcomes for both detection and estimation by harnessing the CSI from every individual measured subcarrier, subsequently consolidating these outcomes. When contrasted against results derived from CSI based on specific extraction protocols or those obtained through weighted summation, the methodology introduced in this study offers substantial improvements in CSI-based detection and estimation performance. This approach not only underscores the significance but also serves as a robust exemplar for the comprehensive application of CSI.

1. Introduction

In the evolving landscape of wireless communication, the fusion of sensing, aptly termed integrated sensing and communication (ISAC), has emerged as a cornerstone for forthcoming wireless paradigms, spanning both short-range and cellular modalities. Wi-Fi, being the preeminent technology for short-range wireless communication, recognizes sensing as a propitious frontier. In alignment with this vision, the IEEE 802.11 Standard Association initiated Project 802.11bf in 2020, envisaging an amendment tailored for WLAN sensing. As a testament to the concerted efforts of its members, the 11bf Task Group (TG) promulgated the 11bf Amendment Draft 2.0 in July 2023. Parallely, in the realm of 6G research, the ITU's Report M.2516–0 underscores that ISAC is poised to be an integral facet of future mobile telecommunication systems [1].

Relative to the receiving signal strength index (RSSI), CSI offers the granularity of channel insights at the subcarrier level. This facilitates the extraction of comprehensive information — encompassing path loss, delay, Doppler effect, and multipath characteristics between transmitter and receiver — which in turn optimally augments various Wi-Fi sensing applications. Over the past decade, a proliferation of research endeavors has leveraged Wi-Fi CSI, with studies delving into realms such as localization [2], breath detection [3], and gesture recognition [4], among others. Furthermore, complex-valued neural network based Wi-Fi sensing pioneer the signal processing method for next-generation Wi-Fi protocols, which is proposed by SLNet [5].

In the domain of breath sensing utilizing CSI subcarriers, a myriad of preprocessing strategies have been introduced to augment sensing efficacy. Bao et al. [6] leveraged a support vector machine to cherry-pick optimal signals from the CSI amplitude and the phase

* Corresponding author.

E-mail address: liuf6@sustech.edu.cn (F. Liu).¹ Co-first authors.<https://doi.org/10.1016/j.jiixd.2024.05.002>

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differential across dual antennas, laying the groundwork for respiratory rate computation. Taking a distinct approach, Chen et al. [7] calculated the time reversal-based resonating strength (TRRS) for each subcarrier's CSI, then invoked eigendecomposition to identify signals imbued with pivotal information for respiratory discernment and estimation. Liu et al. [8] harnessed sinusoidal function fitting on individual subcarrier CSI for respiratory rate projections, subsequently synthesizing these weighted estimations. However, Zhang [9] underscored the potential pitfalls of the sinusoidal fitting approach, particularly its proclivity to yield spurious estimates under specific conditions. Venturing into another avenue, Wang et al. [10] pinpointed CSI data clusters with heightened sensitivity to respiration, gauging signal energy and motion detection. They then applied peak detection for rate computations within these clusters, with the median value representing the definitive estimated rate. Xie et al. [11] employed a principal component analysis (PCA) scheme to cull respiration-aligned principal signal facets from the CSI ratio, subsequently deploying these facets for breath signal detection. Zhuo et al. [12] separately estimated respiratory rates for each subcarrier, culminating in an aggregate rate via weighted summation. Both Zhang [13] and Gu [14] opted for subcarriers evincing maximal variance, subsequently utilizing these selected subcarriers for respiratory rate estimation.

In conclusion, for breath sensing tasks, predominant preprocessing methodologies primarily pivot around the judicious extraction of optimal subcarriers based on prescribed criteria, or the incorporation of methods such as PCA to discern signal facets rich in respiratory intel. Such methodologies are earmarked as the first category of CSI preprocessing in our framework. When it pertains to respiratory rate computations, beyond the tools harnessed for breath detection, a recurrent approach involves the computation of respiratory rates tethered to individual subcarrier CSIs, culminating in a weighted synthesis to deduce the definitive rate. This is categorized under the second tier of our CSI preprocessing taxonomy.

This paper undertakes an exhaustive juxtaposition of these two salient preprocessing paradigms for respiratory sensing. It accentuates the superior efficacy of the second preprocessing category in mining the intricacies nested within subcarrier CSIs. This proficiency is manifested through bolstered accuracy in breath detection and amplified precision in rate estimations. Within this scope, the contributions of our work are summarized as follows:

- We summarize the existing CSI preprocessing methods for sensing and analyze their shortcomings.
- We derive the signal model for CSI-radio based Wi-Fi sensing and propose a voting based algorithmic framework to enhance the performance for Wi-Fi sensing.
- We verify our algorithmic framework using actual measured data. We make comparative analysis between our proposed method and existing methods to explain the accuracy enhancement of the proposed method.

This paper is structured as follows: Section 2 offers a comprehensive overview of conventional sensing mechanisms rooted in CSI. In Section 3, we elaborate on our proposed CSI-centric algorithmic framework, providing a theoretical exposition on the imperative of initially procuring sensing outcomes from individual subcarrier CSIs, followed by a statistical amalgamation of these outcomes. Empirical validation results, derived from respiratory sensing datasets, are showcased in Section 4, underscoring the efficacy of our proposed methodology. Section 5 encapsulates the concluding remarks and charts potential avenues for future research.

2. Preliminary

Fig. 1 delineates the algorithmic framework for CSI-based sensing employing the first category of CSI preprocessing. As depicted in Fig. 1, the CSI ratio [15] is harnessed to mitigate non-ideal synchronization offsets, including carrier frequency offset (CFO), sampling frequency offset (SFO), sampling time offset (STO), and cyclic shift diversity (CSD) [16]. To make full use of all the valid subcarriers,

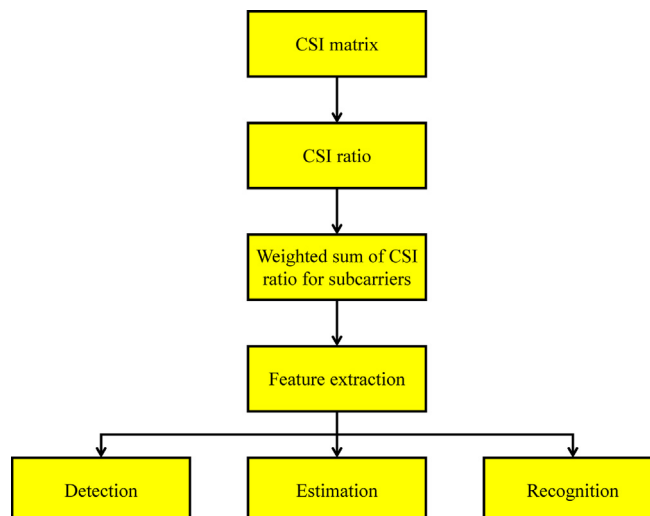


Fig. 1. The conventional sensing algorithmic process based on CSI.

weighted summation is typically employed to aggregate the subcarriers of the CSI. Subsequently, signal processing methods for feature extraction are applied to the “weighted sum of CSI ratio” for Wi-Fi sensing application, including but not limited to human existing detection, respiratory rate estimation, activity recognition.

Within the context of the orthogonal frequency division multiplexing (OFDM) system, individual subcarriers may manifest non-uniform channel responses — captured meticulously by the CSI. It necessitates the assignment of distinct weight factors to each subcarrier in the execution of the weighted summation. These weight factors can be determined based on multiple criteria, ranging from the reliability of the CSI estimate to the relative significance of each subcarrier.

While the “weighted sum of CSI for subcarriers” offers an aggregate CSI perspective, it inherently risks overlooking the nuanced information inherent to individual subcarriers. By consolidating multi-carrier CSI into a singular metric, the intricate details unique to each subcarrier may become obfuscated. Such a diminution in information granularity might compromise performance, particularly in intricate or dynamic channel contexts. Addressing this limitation, this paper introduces and delves into an alternative processing paradigm designed to mitigate potential information attrition at the CSI subcarrier level.

3. Methods

Fig. 2 shows the CSI based sensing algorithmic process procedure proposed in this paper. Feature extraction is conducted at each individual subcarrier and weighted statistical voting is performed based on all the signal subcarrier results.

Assuming we have N subcarriers, let's define:

- CSI_i : The CSI for the i th subcarrier.
- $D(CSI_i)$: The detection result for the i th subcarrier, based on the CSI for that subcarrier.
- $V(D(CSI_1), D(CSI_2), \dots, D(CSI_N))$: The voting function, which combines the detection results for each subcarrier to give the final decision.

The final decision based on statistical voting can be represented as:

$$\text{Decision}_{\text{Voting}} = V(D(CSI_1), D(CSI_2), \dots, D(CSI_N)).$$

The voting-based method offers a distinct advantage: It retains and capitalizes on the granular details inherent in each subcarrier. By determining the outcome for each subcarrier independently and subsequently amalgamating these individual decisions, the methodology can potentially yield a more precise and resilient composite decision, especially in the face of intricate or dynamically changing channel environments. Conversely, the weighted summation approach might overlook the fine-grained details of each subcarrier, as it amalgamates CSI values prior to final decision-making.

The comparative effectiveness of the voting-based method hinges on prevailing channel conditions, the fidelity of the CSI estimates, and the intrinsic attributes of the detection algorithm and voting function. Consequently, a thorough assessment of both methods' performances, achieved either through rigorous simulations or empirical tests, is paramount.

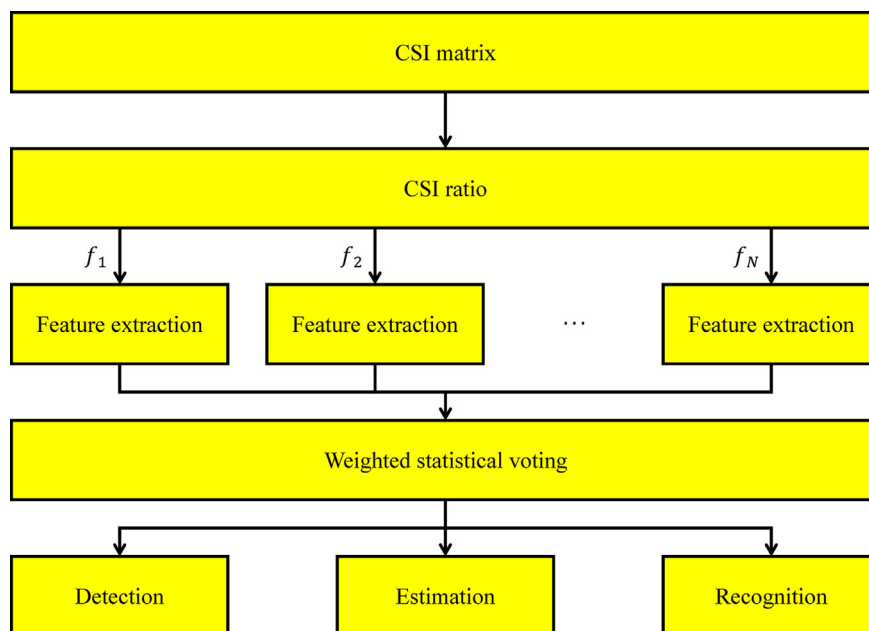


Fig. 2. Proposed sensing algorithmic process based on CSI.

Assuming the frequency offset between the transmitter and receiver is Δf , and utilizing the N -point Fourier transform and inverse Fourier transform, where N_g denotes the size of the cyclic prefix. The size of an OFDM symbol is then given as $N_s = N_g + N$. Let T represent the sampling period of the data and define $\epsilon = \Delta f NT$. The complex data symbol on the k th subcarrier of the l th received OFDM symbol, denoted as $c_{k,l}$, can be expressed as follows [17]:

$$c_{k,l} = \exp(j\pi((N-1)/N)\epsilon) \exp(j2\pi((lN_s + N_g)/N)\epsilon) \frac{\sin \pi\epsilon}{N \sin(\pi\epsilon/N)} H_{k,l} d_{k,l} + ICI_{k,l} + W_{k,l}, \quad (0 \leq k < N, -\infty \leq l < \infty). \quad (1)$$

In the above, $H_{k,l}$ represents the channel transfer function, while $d_{k,l}$ indicates the theoretically mapped data symbols. $ICI_{k,l}$ denotes the inter-carrier interference noise caused by carrier frequency offset, and $W_{k,l}$ corresponds to the interference term from the white noise.

Additionally, the symbols obtained from sampling undergo a symbol synchronization process, which introduces a time delay correction term for $c_{k,l}$. This correction term corresponds to a shift of N_l . When N_l is positive, it represents an advance shift, whereas a negative N_l indicates a delayed shift. Therefore, the symbol used for deriving the CSI at this point is

$$\tilde{c}_{k,l} = \exp(j\pi((N-1)/N)\epsilon) \exp(j2\pi((lN_s + N_g + N_l)/N)\epsilon) \frac{\sin \pi\epsilon}{N \sin(\pi\epsilon/N)} H_{k,l} d_{k,l} + ICI_{k,l} + W_{k,l}, \quad (0 \leq k < N, -\infty \leq l < \infty). \quad (2)$$

From equation (2), it's evident that the interference associated with a single subcarrier consists of three components: the signal distortion directly attributed to the frequency offset Δf between the transmitter and receiver, the inter-carrier interference $ICI_{k,l}$ triggered by the same frequency offset, and the white noise term $W_{k,l}$. While the white noise term $W_{k,l}$ can be suppressed by a linear weighted summation of CSI across subcarriers, the interference stemming from non-white noise terms often isn't adequately addressed by this method, due to their significant correlation across subcarriers. Moreover, interference induced by the frequency offset between transmitting and receiving devices often dominates the error terms. As a result, the traditional CSI preprocessing method of Weighted summation might not be an effective approach for CSI preprocessing.

Set $a_l = \exp(j\pi((N-1)/N)\epsilon) \exp(j2\pi((lN_s + N_g)/N)\epsilon) \frac{\sin \pi\epsilon}{N \sin(\pi\epsilon/N)}$, $b_{k,l} = ICI_{k,l} + W_{k,l}$ then CSI can be expressed as:

$$\tilde{c}_{k,l}/d_{k,l} = \exp\left(j2\pi \frac{N_l}{N} \epsilon\right) a_l H_{k,l} + b_{k,l}/d_{k,l}, \quad (0 \leq k < N, -\infty \leq l < \infty). \quad (3)$$

Furthermore, we can derive the functional expression of the CSI ratio, denoted as $r_{k,l}$, which is as follows:

$$r_{k,l} = \exp\left(j2\pi \frac{\Delta N}{N} \epsilon\right) H_{k,l}^d + V_{k,l}, \quad (0 \leq k < N, -\infty \leq l < \infty), \quad (4)$$

in which $V_{k,l} = \left(b_{k,l}/d_{k,l} - b'_{k,l}/d_{k,l} \exp(j2\pi \frac{\Delta N}{N} \epsilon) H_{k,l}^d\right) / \left(\exp\left(j2\pi \frac{N'_l}{N} \epsilon\right) a_l H'_{k,l} + b'_{k,l}/d_{k,l}\right)$.

Referring to equation (4), the presence or absence of the superscript ' distinguishes between antenna 1 and antenna 2 at the receiver, respectively. Given the nuanced differences in signals received by the two antennas, the shift values N_l and N'_l , ascertained from individual symbol synchronization of their respective received data, might diverge. Consequently, $\Delta N = N_l - N'_l$ could emerge as a random value, not necessarily equating to zero. Furthermore, $H_{k,l}^d = \frac{H_{k,l}}{H_{k,l}}$.

Evaluation of equation (4) suggests that the CSI ratio employed for respiratory sensing integrates multiple noise facets. Notably, an error, attributable to the non-aligned shift values of symbol synchronization, forms part of the multiplicative noise. Additionally, the additive noise term $V_{k,l}$ is conspicuously non-Gaussian white noise.

Next, the effect of linear weighting on the signal is analyzed. The definition of linear weighting is

$$R_l = \sum_{k=1}^N w_k * r_{k,l} = \underbrace{\exp\left(j2\pi \frac{\Delta N}{N} \epsilon\right) \sum_{k=1}^N w_k * H_{k,l}^d}_{\text{I}} + \underbrace{\sum_{k=1}^N w_k * V_{k,l}}_{\text{II}}, \quad (0 \leq k < N, -\infty \leq l < \infty). \quad (5)$$

In part I, the multiplicative noise term $\exp(j2\pi \frac{\Delta N}{N} \epsilon)$ still exists. A weighted sum of $H_{k,l}^d$ does not necessarily result in better signal quality than a single $H_{k,l}^d$.

For part II, it is expressed as:

$$\begin{aligned}
\sum_{k=1}^N w_k * V_{k,l} &= \sum_{k=1}^N w_k * \frac{\left(b_{k,l} - b'_{k,l} \exp\left(j2\pi \frac{\Delta N}{N} \epsilon\right) H_{k,l}^d \right)}{\left(\exp\left(j2\pi \frac{N_l}{N} \epsilon\right) a_l H'_{k,l} d_{k,l} + b'_{k,l} \right)} = \\
&= \sum_{k=1}^N w_k * \frac{\left(\text{ICI}_{k,l} + W_{k,l} - \left(\text{ICI}'_{k,l} + W'_{k,l} \right) \exp\left(j2\pi \frac{\Delta N}{N} \epsilon\right) H_{k,l}^d \right)}{\left(\exp\left(j2\pi \frac{N_l}{N} \epsilon\right) a_l H'_{k,l} d_{k,l} + \text{ICI}'_{k,l} + W'_{k,l} \right)} \approx \\
&= \sum_{k=1}^N w_k * \frac{\left(\text{ICI}_{k,l} - \text{ICI}'_{k,l} \exp\left(j2\pi \frac{\Delta N}{N} \epsilon\right) H_{k,l}^d \right) + W_{k,l} - W'_{k,l} \exp\left(j2\pi \frac{\Delta N}{N} \epsilon\right) H_{k,l}^d}{\left(\exp\left(j2\pi \frac{N_l}{N} \epsilon\right) a_l H'_{k,l} d_{k,l} + \text{ICI}'_{k,l} \right)}. \quad (6)
\end{aligned}$$

In OFDM systems, ICI primarily originates from elements like frequency offset and Doppler frequency offset. Furthermore, the denominator term in equation (6) corresponds directly to the symbol $\tilde{c}_{k,l}$, and the influence of $W'_{k,l}$ on this term can be neglected. Considering that the white Gaussian noise term in the numerator is significantly suppressed after weighted summation, equation (6) can be further simplified as:

$$\sum_{k=1}^N w_k * V_{k,l} \approx \sum_{k=1}^N w_k * \frac{\left(\text{ICI}_{k,l} - \text{ICI}'_{k,l} \exp\left(j2\pi \frac{\Delta N}{N} \epsilon\right) H_{k,l}^d \right)}{\left(\exp\left(j2\pi \frac{N_l}{N} \epsilon\right) a_l H'_{k,l} d_{k,l} + \text{ICI}'_{k,l} \right)}. \quad (7)$$

Upon a more intricate examination of equation (7), it is discerned that the primary sources of ICI error arise from factors like the frequency discrepancies between the transceiver devices and Doppler frequency shifts. This underpins the strong correlation observed amongst the $\text{ICI}_{k,l}$ values or $\text{ICI}'_{k,l}$ values across the subcarriers. Concurrently, $H_{k,l}^d$ corresponding to individual subcarriers exhibits a pronounced correlation. A notable characteristic of the components in equation (7) is their strong correlation. The foundational assumption for efficacious noise suppression through linear weighting is the independent distribution of noise. Given this scenario, employing a linear weighting scheme may not necessarily mitigate the respective noise component present in equation (7).

This in-depth examination further informs the sensing strategy elucidated in Fig. 2.

4. Actual data processing result and analysis

4.1. Improvement in detection performance by the new method

As one of the most important Wi-Fi sensing applications, breath detection is adopted to evaluate the performance of the proposed signal processing procedure.

802.11n Wi-Fi system is used to collect the CSI when a person remained stationary in a room. 52 subcarriers with 312.5 kHz subcarrier spacing CSI is collected in this environment. Fig. 3 shows an example of test scenarios.

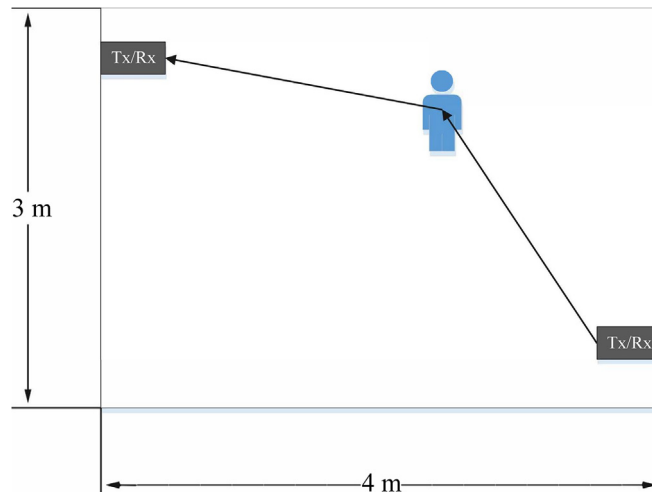


Fig. 3. Example of scenario for obtaining measured CSI.

To benchmark the efficacy of the signal processing method delineated in this study, a uniform dataset is subjected to disparate processing procedures. Specifically, for the respiratory detection algorithm depicted in Fig. 1, we employed three prevalent linear weighting strategies: equal weighting of every subcarrier's CSI, selecting the subcarrier CSI that corresponds to the maximum sum of absolute values, and optimal subcarrier CSI selection via PCA.

This study introduces a detector grounded in time-frequency analysis, characterized by a markedly low false alarm rate. Drawing from prior empirical observations, the detector's false alarm rate typically remains below 0.3. Given this backdrop, it's feasible to statistically integrate detection results predicated on individual subcarriers. Consequently, the fusion criteria for breath sensing are defined as: Given M subcarriers, if the probability of accurate detection pertaining to the k th subcarrier is p_k , and the number of human presence N_0 in all detection results is larger than or equal to $0.3M$, then the final detection result can be identified as human presence.

Given the variability in the signal quality of the CSI across different subcarriers, one can postulate that the probability of accurately detecting human presence based on the CSI from individual subcarriers follows distinct and independent distributions. For illustrative clarity, only three subcarriers are utilized for a numerical simulation to provide an intuitive understanding of the fusion detection outcomes from a statistical standpoint. In this simulation, the probability of correct human presence detection leveraging the CSI from each subcarrier is hypothesized to uniformly distribute within the interval $[0, 0.8]$. The outcomes of 20 such simulation experiments are depicted in Fig. 4. A similar enhancement in detection performance is observed for a system with 52 subcarriers.

The processing results are shown in Fig. 5. As can be seen from Fig. 2, by utilizing the algorithmic process proposed in this paper, the false alarm probability can be significantly reduced, even though it may lead to a slight increase in the permissible error alarm probability.

4.2. Improvement in estimation performance by the new method

This study utilizes sampled human respiratory data for experimental validation. The experiments were conducted with a single individual breathing in the room. A total of 30 sets of experimental data were collected and analyzed. It's important to note that in these experiments, the number of subcarriers was set to 30. The results of the experimental tests are presented in Figs. 6 and 7.

In Fig. 6, it can be seen that fused estimation results of all subcarriers have significantly higher estimation accuracy than that of each single subcarrier. In addition, Fig. 7 also shows that the accuracy of respiration rate estimation obtained by fusing the sensing result of each subcarrier is higher than that of respiration rate estimation obtained by using linear weighting preprocessing for the CSI. In the estimation experiment, for simplicity, the estimation strategy used is described as follows: first, the CSI ratio frequency is measured using PLL [18], then the spectrum of the measured frequency is calculated. Afterward, within the possible range of respiration, the frequency point corresponding to the maximum amplitude value is searched to calculate the estimated value of the respiration rate.

4.3. Comparative analysis and lessons learned from the results

Channel state information (CSI) holds significant importance in OFDM systems and is the most widely used information in Wi-Fi sensing technology. Within an OFDM framework, comprising multiple subcarriers, the CSI matrix encapsulates vital information spanning both time and frequency domains. The performance of Wi-Fi sensing heavily relies on the method to deal with the subcarriers in different frequency domain.

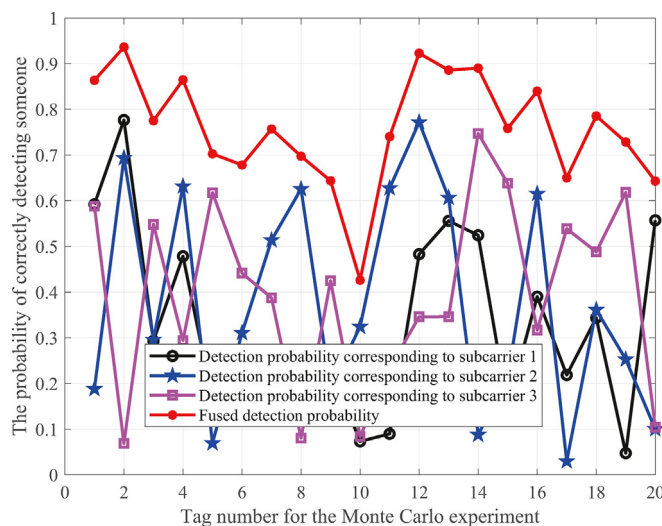


Fig. 4. The simulation shows the correct detection probability corresponding to the data of three subcarriers, and also shows the detection probability after fusion according to the definition of fusion sensing.

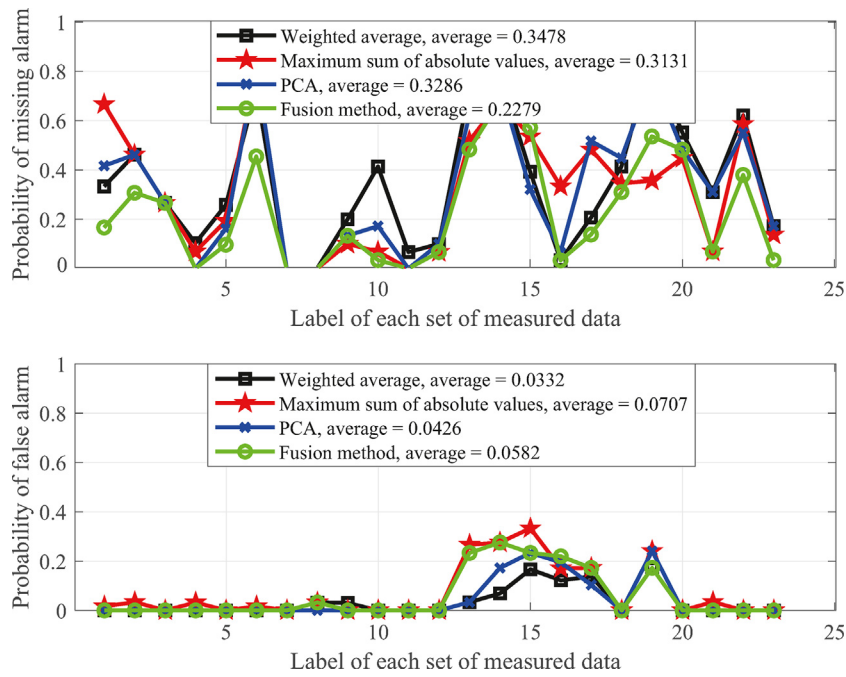


Fig. 5. The black line, red line and blue line represent the breathing detection results obtained by using the algorithmic process illustrated in Fig. 1, corresponding to three different CSI linear weighting methods: the equal weight addition of each subcarrier's CSI, the subcarrier CSI corresponding to the maximum sum of absolute values, and the optimal subcarrier CSI is selected by PCA respectively. The green line represents the breathing detection result derived from the algorithmic process associated with Fig. 2.

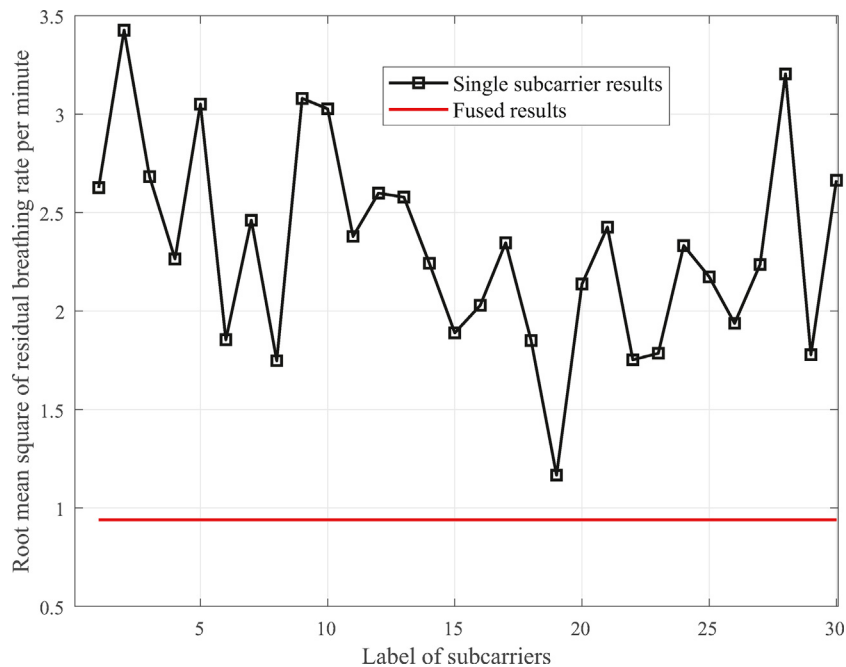


Fig. 6. The black line represents the root mean square error of the residuals from the estimated results, obtained by separately processing the CSI data of each subcarrier according to the algorithmic process shown in Fig. 1. The red line indicates the root mean square error of the residuals from the estimated results derived from the algorithmic process depicted in Fig. 2. The graph demonstrates that fusing the sensing results from each subcarrier's CSI can yield a higher estimation accuracy than utilizing the sensing result from any single subcarrier's CSI individually.

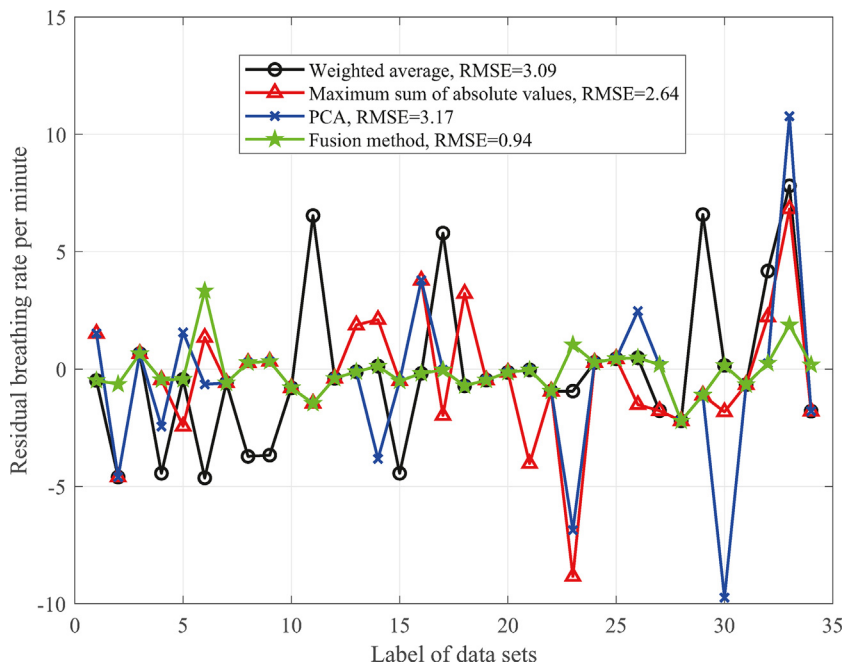


Fig. 7. The black line, red line and blue line correspond to the root mean square errors of the respiration rate estimates obtained through the algorithmic process depicted in Fig. 1. Specifically, the CSI linear weighting modes they correspond to are the equal-weighted summation of subcarrier CSIs for the black line, the subcarrier CSI corresponding to the maximum sum of absolute values for the red line, and the optimal subcarrier CSI is selected by PCA for the blue line. The green line signifies the root mean square error of the respiration rate estimate obtained through the algorithmic process represented in Fig. 2. It can be observed that the sensing algorithm based on the algorithmic process shown in Fig. 2 significantly yields higher precision in the estimated results.

In comparison to existing subcarrier selection methods, our proposed “voting method” demonstrates superior performance in terms of both the probability of missing alarm and root mean square (RMS) when applied to actual measured data, which may be explained as follows. The CSI matrix is multidimensional, exhibiting a tendency for consistency over time in the time domain. However, at a fixed time index, CSI can vary in the frequency domain due to the intricate information embedded within each subcarrier. On one hand, methods that involve selection among subcarriers (such as optimal subcarrier selection via PCA) may inadvertently discard valuable information contained in other subcarriers. On the other hand, approaches like the “weighted sum of CSI for subcarriers” method, which consolidate multi-carrier CSI into a single subcarrier CSI, capitalize on the entirety of subcarriers. However, these preprocessing methods may obscure the intricate details inherent in individual subcarriers.

Our proposed “voting method” not only maximizes the utilization of information from valid subcarriers but also preserves intricate details, rendering it more reliable and greatly enhancing its robustness for adapting to various environments.

5. Conclusion

In this work, we theoretically analyze the problems present in the CSI based sensing, revealing the inadequacies of traditional CSI preprocessing methods for Wi-Fi sensing. Subsequently, we propose a “voting based method” that enables comprehensive utilization of the information contained in each subcarrier. To verify the efficiency and reliability of our proposed method, we compare the detection and estimation performance of our proposed method with the existing methods using actual measured data. An enhanced sensing performance is achieved through the statistically fused sensing result corresponding to each subcarrier's CSI. Although our method is only applied to respiratory estimation application in this work, it offers a novel algorithmic framework for high-performance Wi-Fi sensing, and for future work, we will do more research on high-performance algorithmic framework for other Wi-Fi sensing applications.

CRedit authorship contribution statement

Tao Deng: Formal analysis, Validation, Writing – original draft. **Bowen Zheng:** Conceptualization, Writing – review & editing. **Rui Du:** Data curation, Formal analysis. **Fan Liu:** Conceptualization, Formal analysis. **Tony Xiao Han:** Project administration, Resources.

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Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Tao Deng graduated with a bachelor's degree from South China University of Technology, and obtained his doctoral degree from the University of Chinese Academy of Sciences/Shanghai Astronomical Observatory, Chinese Academy of Sciences. He conducted postdoctoral research at Huawei for two years and is now an assistant researcher at a national laboratory. His main research area is related to astrometry.



Bowen Zheng is a Master student at the Southern University of Science and Technology, Shenzhen 518055, China. He is interested in integrated sensing and communication (ISAC), Wi-Fi sensing and radar signal processing.



Rui Du received the M.S. and Ph.D. degrees in Information and Communication Engineering from Northwestern Polytechnical University, in 2014 and 2018 respectively. During 2015–2017, he was a visiting PhD student at Microwave Integrated Systems Laboratory, University of Birmingham. He joined Wireless Technology Laboratory, Huawei, 2018. His current research focuses on wireless communication, signal processing, integrated communication and sensing (ISAC) and standardization of wireless communication.



Fan Liu received his Ph.D. degree from Beijing Institute of Technology, China, in 2018. He is now an assistant professor at the Southern University of Science and Technology, Shenzhen 518055, China. Prior to that, he was a Marie Curie Research Fellow at the University College London, from 2018 to 2020. He serves as the founding academic chair of the IEEE Communications Society (ComSoc) Integrated Sensing and Communications (ISAC) Emerging Technology Initiative, an associate editor of IEEE Communications Letters and IEEE Open Journal of Signal Processing, and a founding member of the IEEE Signal Processing Society (SPS) ISAC Technical Working Group. He was a recipient of the 2023 ComSoc Stephan O. Rice Prize, 2023 IEEE ICC Best Paper Award, 2021 SPS Young Author Best Paper Award, and 2019 Chinese Institute of Electronics Best Doctoral Thesis Award. His research interests lie in the general area of signal processing, wireless communications, and in particular in the field of ISAC. He is a Senior Member of the IEEE.



Tony Xiao Han (Senior Member, IEEE) received the B.E. degree in electrical engineering from Sichuan University, Chengdu, China, and the Ph.D. degree in communications engineering from Zhejiang University, Hangzhou, China. He is currently a Research Expert and Project Manager with Huawei Technologies Company, Ltd., Shenzhen, China. He was a Postdoctoral Research Fellow with the National University of Singapore, Singapore. His research interests include wireless communications, signal processing, integrated sensing and communication (ISAC), and standardization of wireless communication. Dr. Han was the Chair of IEEE 802.11 WLAN Sensing Topic Interest Group and Chair of 802.11 WLAN Sensing Study Group. He is the Chair of IEEE 802.11bf WLAN Sensing Task Group. He is also the Founding Industry Chair of IEEE ComSoc ISAC Emerging Technology Initiative (ISAC-ETI), Vice Chair of IEEE WTC Special Interest Group on ISAC, Guest Editor of the IEEE Journal on Selected Areas in Communications Special Issue on Integrated Sensing and Communications (ISAC). He was the Co-Chair of many workshops, such as IEEE GLOBECOM 2020 workshop on ISAC.